Group 88 - UDL Assignment 2: Part C

Comparative Analysis: beta-VAE, VQ-VAE + PixelCNN, WGAN and WGAN-GP

CIFAR-10 | PyTorch | Standard Inception FID on 5,000 real and generated images

Evaluation summary

Model	Reconstruction PSNR (dB)	Standard FID ↓	Training (min)
beta-VAE (β=1)	18.02	200.98	4.64
VQ-VAE + PixelCNN (K=256)	24.20	150.42	12.15
WGAN	N/A	168.33	8.31
WGAN-GP	N/A	158.47	14.08

Comparative discussion

Reconstruction error. PSNR is meaningful only where an input image is reconstructed. VQ-VAE (K=256) achieved 24.20 dB, clearly above beta-VAE (β=1) at 18.02 dB, so it produced lower pixel reconstruction error. Among all VQ codebooks, K=512 had the best reconstruction PSNR (24.77 dB); K=256 is retained here because the assignment specifies it for the Part C visual comparison.

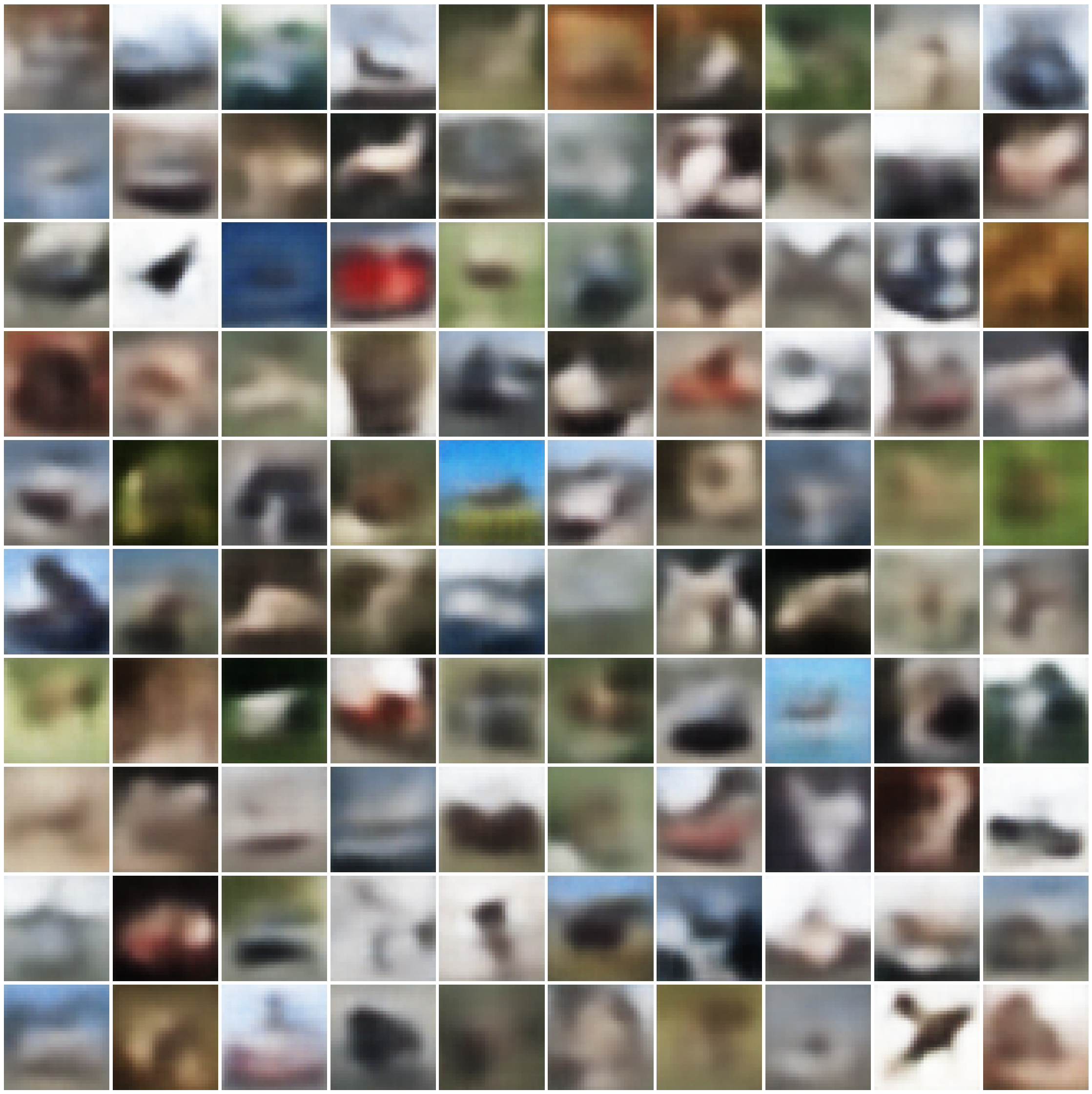
Sampling quality. Standard Inception FID compares distributions of unpaired real and generated images, so it is appropriate for every model. VQ-VAE + PixelCNN (K=256) had the best Part C FID (150.42). WGAN-GP improved on WGAN (158.47 versus 168.33), supporting the expected benefit of gradient penalty over weight clipping. For beta-VAE, increasing beta degraded FID: beta=1 was 200.98, beta=2 was 244.29, beta=4 was 276.73, and beta=10 was 313.46. Within the VQ sweep, K=128 had the lowest FID (139.33), showing that the FID/reconstruction trade-off should be discussed rather than assuming the highest PSNR setting is best for samples.

Speed. beta-VAE was fastest at 4.64 minutes per run because it trains one encoder-decoder model. VQ-VAE + PixelCNN took 12.15 minutes because it trains the VQ-VAE, extracts discrete latents, and trains an autoregressive prior. WGAN took 8.31 minutes; WGAN-GP was slowest at 14.08 minutes because its gradient penalty adds differentiation work during critic updates.

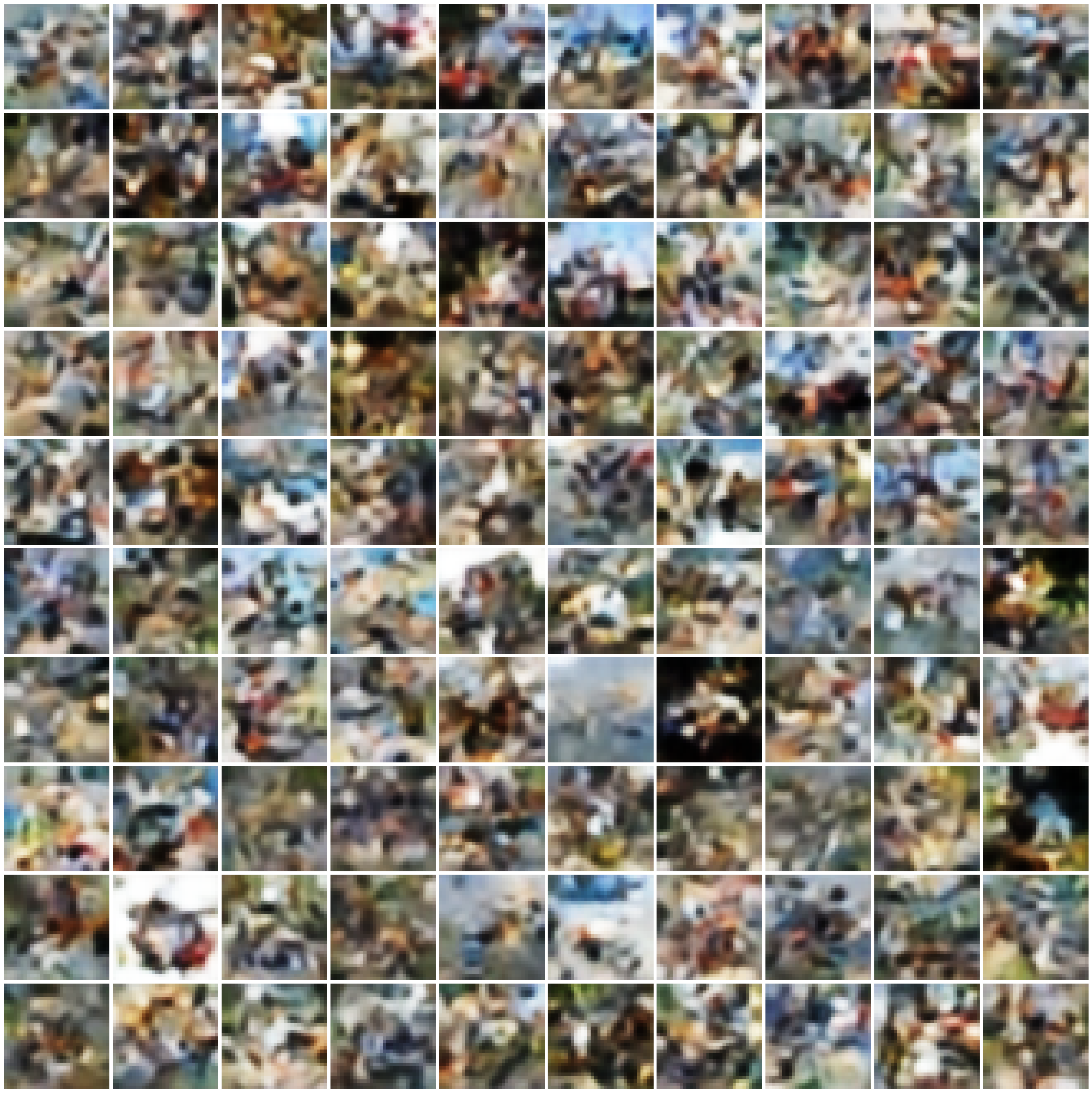
Conclusion. VQ-VAE gave the strongest reconstruction result for the required K=256 comparison and the best Part C FID. WGAN-GP was the stronger GAN variant, at a higher compute cost. PSNR is deliberately marked N/A for GANs because they have no paired reconstruction target.

100 visual examples per model

beta-VAE (β=1) reconstructions



VQ-VAE + PixelCNN (K=256) samples



WGAN samples

WGAN-GP samples